Curvature Components and Restricted Torsion Variations in Einstein–Cartan Minisuperspace

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Abstract

We compare Einstein–Cartan actions with LC-Weyl, irreducible EC-Weyl, and Ricci-tracefree EC quadratic invariants on compact homogeneous spatial geometries times a circle. At fixed geometry, circle-dependent axial and vector-trace torsion leaves irreducible EC-Weyl equal to LC-Weyl; the additional nine-component and four-form sectors supply the axial gradient term. For the Ricci-tracefree action we classify all smooth periodic stationary torsion profiles: nonpositive coupling admits only zero torsion, while positive coupling admits zero, resonant harmonic, and constant mixed branches, all restricted saddles. Constant relative torsion-axis orientations have extrema and degeneracies determined by spatial anisotropy. For a coupled periodic shape–twist family on $S^3 \times S^1$, the nine-component sector contributes a further quadratic frequency term, while all three actions share a strictly positive quartic curvature coefficient. At fixed amplitude their high-frequency actions grow for negative coupling and decrease without bound for positive coupling; finite-frequency configurations below the round reference can remain. Five normalized static shape directions exhibit the nonstationary Nil reference and its flat paths, whose fixed compact quotients can be distinguished by an integral harmonic-form invariant. These are results for specified Euclidean variation families, with the static identities and classifications of the preceding study used as reference results.

1 Introduction

Quadratic curvature terms in Einstein–Cartan gravity can probe different parts of torsionful curvature. Removing the Ricci traces leaves an irreducible Weyl tensor together with a nine-component pair-antisymmetric sector and a one-component four-form sector. Their distinct dependence on torsion raises a variational question: which components contribute when the torsion amplitude or its orientation changes? We study this question on $S^3 \times S^1$, $T^3 \times S^1$, and $\mathrm{Nil}^3 \times S^1$.

Papers I–III provide the geometric conventions and related variational results [1, 2, 3]. Paper I classifies radius potentials with geometric torsion and the complete Nieh–Yan term. Paper II establishes the curvature decomposition, the static aligned-torsion identities, and the radius-and-shape potential bounds. Paper III develops the scalar, twist, and shape sectors of the EC action at zero quadratic-curvature coupling. Building on these results, we compare actions with LC-Weyl, irreducible EC-Weyl, and Ricci-tracefree EC quadratic invariants. All three retain the same EC scalar term. The static classification of Paper II supplies the reference limit; the focus here is the derivative and orientation dependence of its curvature components.

The first result is an exact separation for circle-dependent torsion at fixed spatial geometry. With axial amplitude $h = \eta/r$ and vector-trace amplitude V, the two additional squared norms are $6(\dot{h} + 2hV/3)^2$ and $6(\dot{h} - 2hV/3)^2$, while the irreducible Weyl tensor equals the LC Weyl tensor. This gives the restricted torsion equations and their periodic-profile consequences. The static analysis uses five independent normalized shape directions and exhibits the nonzero Nil shape gradient and its two exactly flat paths. A further component-specific effect appears under rotations of the torsion axis: on anisotropic geometry, the additional nine-component sector supplies an orientation-dependent term and mixed cubic interactions.

For the Ricci-tracefree action at positive coupling we also classify all smooth periodic stationary torsion profiles, including the algebraically exceptional mixed branch, and show that each is a restricted saddle. The constant-axis formula gives global finite-angle extrema and their degeneracies. A coupled shape–twist family then extends the component comparison to varying geometry: its additional nine-component term is quadratic in frequency, whereas a common strictly positive quartic curvature coefficient controls the high-frequency behavior of all three actions. On Nil, an integral harmonic-form invariant distinguishes compact metrics along the potential-flat paths.

These variation families are kept distinct. Circle torsion in Section 3 has fixed geometry; Sections 4–5 treat separate static variations; Section 6 retains metric derivatives on a specified coupled periodic family at fixed scalar controls. Section 2 gives their definitions and Section 7 discusses their scope. The appendices provide projections, derivative calculations, coordinate maps, reproduction instructions and the revision note, followed by the fixed-quotient argument.

2 Geometry, actions, and variation families

2.1 Coframe and torsion

Use an oriented Euclidean orthonormal frame, spatial indices $i, j, k = 0, 1, 2$, and circle index 3. The circle coordinate has period one and L is its physical circumference. At zero twist and mixing, the coframe is

$$e^i = rM^i{}_j\sigma^j, \qquad e^3 = L\,\mathrm{d}y, \qquad \det M = 1, \quad y \sim y + 1, \quad r, L > 0. \tag{G-01}$$

For S^3, use $\mathrm{d}\sigma^i = 4\sigma^j \wedge \sigma^k$ cyclically; the round physical radius is $r/2$. The torus forms are closed and the three angular periods are 2π. For Nil, take $\sigma^0 = \mathrm{d}x$, $\sigma^1 = \mathrm{d}w$, $\sigma^2 = \mathrm{d}z - x\mathrm{d}w$, with periods $(2\pi, 2\pi, 4\pi^2)$ and the x-face gluing $z \mapsto z + 2\pi w$. These identifications are held fixed under variations. Thus

$$\mathrm{Vol} = c_M L r^3, \qquad c_{S^3} = \frac{\pi^2}{4}, \quad c_{T^3} = (2\pi)^3, \quad c_{\mathrm{Nil}^3} = (2\pi)^4. \tag{G-02}$$

The torsion is geometric torsion, defined by the first Cartan equation, with aligned axial and vector-trace parts

$$\begin{aligned} T^a &= \mathrm{d}e^a + \omega^a{}_b \wedge e^b = \frac{1}{2}T^a{}_{bc}e^b \wedge e^c, \qquad h = \frac{\eta}{r}, \\ T^{\mathrm{AX}}_{ijk} &= 2h\varepsilon_{ijk}, \qquad T^{\mathrm{VT}}_{abc} = \frac{V}{3}(\delta_{ac}\delta_{b3} - \delta_{ab}\delta_{c3}). \end{aligned} \tag{G-03}$$

AX means $V = 0$, VT means $h = 0$, and MX allows both. The supplement accepts geometric torsion directly. The signs and index order of the connection are fixed by

$$\begin{aligned} \mathrm{d}e^a &= \frac{1}{2} C^a{}_{bc} e^b \wedge e^c, \qquad \omega_{ab} = \Gamma_{abc} e^c, \\ \Gamma^{\mathrm{LC}}_{abc} &= \frac{1}{2}(C_{abc} + C_{cba} - C_{bac}), \\ K_{abc} &= -\frac{1}{2}(T_{abc} + T_{bca} - T_{cab}), \qquad \Gamma = \Gamma^{\mathrm{LC}} + K. \end{aligned} \tag{G-04}$$

2.2 Three quadratic invariants

Write W_{EC} for the irreducible Weyl tensor and $\widehat{R}_{\mathrm{EC}}$ for the Ricci-tracefree tensor. Appendix A defines their projections without assuming pair exchange or the algebraic torsion-free Bianchi identity. In four Euclidean dimensions,

$$\begin{aligned} \widehat{R}_{\mathrm{EC}} &= W_{\mathrm{EC}} + R^{(2)} + R^{(3)}, \\ I_{\mathrm{tf}} &= I_{\mathrm{W}} + I_2 + I_3, \qquad I_{\mathrm{W}} = \|W_{\mathrm{EC}}\|^2, \quad I_a = \|R^{(a)}\|^2 \ (a = 2, 3). \end{aligned} \tag{C-01}$$

The component dimensions are $10 + 9 + 1$. $R^{(2)}$ is pair-antisymmetric and Ricci-free; $R^{(3)}$ is a four-form. The LC comparison invariant is $I_{\mathrm{LC}} = \|W_{\mathrm{LC}}\|^2$. The three actions use $X \in \{\mathrm{LC}, \mathrm{W}, \mathrm{tf}\}$ in

$$\begin{aligned} S_X &= -\int \mathrm{vol} \left[\frac{R_{\mathrm{EC}}}{2\kappa^2} + \alpha I_X + \theta \mathcal{N} \right], \qquad \kappa^2 > 0, \\ \mathcal{N}\,\mathrm{vol} &= T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab} = \mathrm{d}(e^a \wedge T_a). \end{aligned} \tag{C-02}$$

The minus sign agrees with Paper III's reduced potential convention. No cosmological term or source is added. α and θ are fixed constants. The static potential U_X is the value of this action on a constant profile.

2.3 What is varied

Family	**Fixed quantities and variations**
Circle torsion	Fix r, L, M and set twist/mixing to zero; vary smooth periodic $\eta(y), V(y)$.
Static shape	Fix r, L, η, V; vary $M = \exp(\sum_A q_A T_A)$ with $\det M = 1$.
Static twist	Fix the spatial geometry and torsion components; replace e^3 by $L(\mathrm{d}y + \omega_i \sigma^i)$.
Static mixing	Set twist to zero; rotate the coframe and respecify the same torsion components in it.
Coupled periodic shape–twist	On S^3, fix r, L, η, V and the torsion axis; vary $M = \mathrm{diag}(f, f^{-1}, 1)$ and $\omega = (w, 0, 0)$ as in Section 6.

The static-mixing prescription changes the torsion axis relative to the metric geometry. It is different from simultaneously rotating all fields as a frame gauge transformation. Radius derivatives below hold η, V fixed, so $h = \eta/r$ changes with r. Derivatives with respect to L are displayed as diagnostics when L was held fixed in finding a radial critical point.

3 Circle-dependent torsion and its curvature components

3.1 Exact derivative-level separation

For constant M, the spatial structure can be diagonalized by a constant spatial orthogonal rotation. Write $\mathrm{d}e^i = n_i e^j \wedge e^k$ cyclically, with constant n_i. Round S^3, flat T^3, and the undeformed Nil coframe correspond to $(4/r, 4/r, 4/r)$, $(0,0,0)$, and $(0,0,-1/r)$. Let $\dot{f} = D_3 f = L^{-1}\mathrm{d}f/\mathrm{d}y$. Direct use of $\mathrm{d}\omega + \omega \wedge \omega$, including $\mathrm{d}K$, gives

$$\begin{aligned} W_{\mathrm{EC}} &= W_{\mathrm{LC}}, \\ I_2 &= 6\left(\dot{h} + \frac{2}{3}hV\right)^2, \qquad I_3 = 6\left(\dot{h} - \frac{2}{3}hV\right)^2, \\ I_{\mathrm{tf}} &= I_{\mathrm{LC}} + 12\dot{h}^2 + \frac{16}{3}h^2V^2, \\ R_{\mathrm{EC}} &= R_{\mathrm{LC}} - 6h^2 - \frac{2}{3}V^2 + 2\dot{V}. \end{aligned} \tag{D-01}$$

These identities hold for arbitrary constant n_i and arbitrary first jets of the two circle profiles. A constant spatial shape changes R_{LC} and I_{LC} while leaving the displayed torsion terms unchanged. Appendix B gives the connection and projected tensor entries. The vector-trace derivative $\dot{V}$ enters the scalar as a total derivative; the Ricci-tracefree quadratic invariant depends on $\dot{h}$.

Torsion restriction	**Extra sectors beyond I_{LC}**
AX: $V = 0$	$I_2 = I_3 = 6\dot{h}^2$; a nonconstant axial profile has a quadratic-gradient term in the trace-free model.
VT: $h = 0$	$I_2 = I_3 = 0$, including a varying V on this fixed geometry.
MX	I_2 and I_3 differ locally; their mixed derivative terms cancel in their equally weighted sum.

At zero derivative, $I_2 = I_3 = 8h^2V^2/3$, recovering Paper II. In the pure AX circle control, the sum is $12\dot{\eta}^2/r^2$, with six units from each extra sector and none from irreducible Weyl. The related local first-jet check for a pure axial field gives $I_{\mathrm{tf}} - I_{\mathrm{LC}} = 4\sum_{i=0}^{2}(D_i h)^2 + 12(D_3 h)^2$. This compares directional coefficients at a point; the actions below vary the torsion only along the circle.

3.2 Restricted action and Euler equations

Let $\chi_X = 1$ for the Ricci-tracefree model and $\chi_X = 0$ for LC-Weyl and irreducible EC-Weyl. After integrating the periodic total derivatives, the exact circle action is

$$\begin{aligned} S_X = c_M L r^3 \int_0^1 \mathrm{d}y \Big[&- \frac{R_{\mathrm{LC}}}{2\kappa^2} - \alpha I_{\mathrm{LC}} + \frac{3h^2}{\kappa^2} + \frac{V^2}{3\kappa^2} \\ &- \chi_X \alpha \left(12\dot{h}^2 + \frac{16}{3}h^2V^2\right)\Big]. \end{aligned} \tag{D-02}$$

Thus the LC and irreducible-Weyl comparisons have algebraic torsion equations $h = V = 0$ in this family. For the Ricci-tracefree model they become

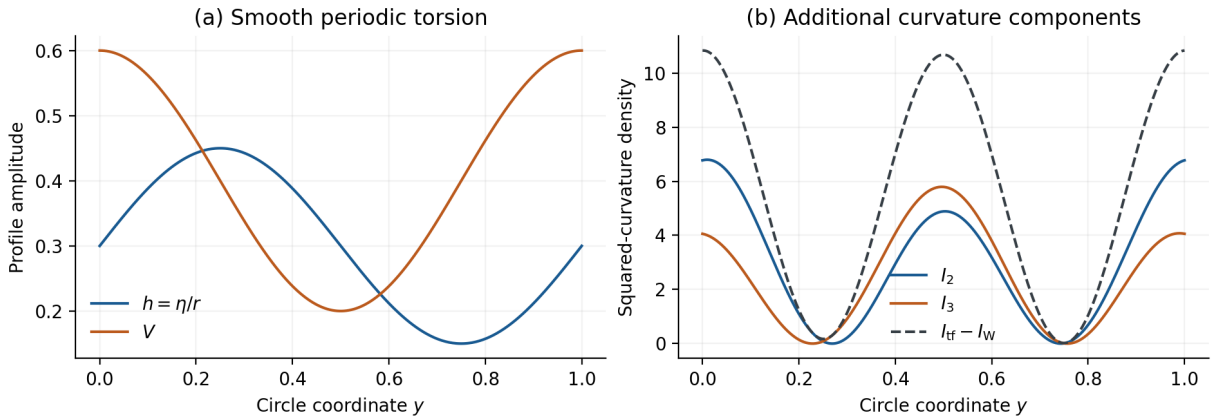


Figure 1: A smooth periodic example on round S^3 with $r = 2$, $L = 1$. The prescribed functions are $h = 0.30 + 0.15\sin(2\pi y)$ and $V = 0.40 + 0.20\cos(2\pi y)$. The two extra curvature norms differ locally; their sum is the Ricci-tracefree correction. $I_{\rm W} = I_{\rm LC} = 0$ on this example. The profiles are prescribed (off shell).

$$\begin{aligned} 24\alpha\ddot{h} + \left(\frac{6}{\kappa^2} - \frac{32\alpha}{3}V^2\right)h = 0, \\ \left(\frac{2}{3\kappa^2} - \frac{32\alpha}{3}h^2\right)V = 0. \end{aligned} \tag{D-03}$$

The velocity Hessian of the bulk density is -24α in the h direction and zero in the V direction. The highest derivative of h is present for $\alpha \neq 0$ even about a nonzero background. The algebraic coefficient of V can vanish; dividing by it would discard branches.

There are two useful consequences within the stated restriction. For $\alpha \leq 0$, multiplying the first Euler equation by h, integrating over the circle, and integrating by parts gives a sum of nonnegative terms:

$$\int_0^1 \mathrm{d}y \left[\frac{6h^2}{\kappa^2} - \frac{32\alpha}{3}h^2V^2 - 24\alpha\dot{h}^2\right] = 0. \tag{D-04}$$

Hence every smooth periodic stationary torsion profile at fixed geometry is $h = V = 0$ for $\alpha \leq 0$. For $\alpha > 0$, the family $h = a\cos(2\pi n y)$, $V = 0$, with nonzero fixed a, yields

$$\frac{S_{\rm tf}[h,0] - S_{\rm tf}[0,0]}{c_M L r^3 a^2} = \frac{3}{2\kappa^2} - \frac{24\pi^2\alpha n^2}{L^2}. \tag{D-05}$$

It tends to minus infinity as the integer frequency increases. This supplies a derivative-driven unbounded direction already at fixed geometry. Its mechanism is additional to the static bounds of Paper II; it is absent from the torsion part of the other two restricted actions.

3.3 Local densities and periodic integrals

$$\begin{aligned} \frac{T^a \wedge T_a}{\mathrm{vol}} &= -4Vh, \qquad \frac{e^a \wedge e^b \wedge R_{ab}}{\mathrm{vol}} = -4Vh + 6\dot{h}, \\ \mathcal{N} &= -6\dot{h}, \\ \mathcal{P} &= \frac{\sum_{a,b} R_{ab} \wedge R_{ab}}{\mathrm{vol}} = D_3\left(2R_{\rm LC}h - 4h^3 - \frac{4}{3}hV^2\right). \end{aligned} \tag{D-06}$$

The curvature–coframe contraction has component expression $\frac{1}{2}\varepsilon^{abcd}R_{abcd}$. The local Nieh–Yan and Pontryagin densities can be nonzero, while both integrals vanish for these smooth periodic profiles with constant geometry. Here $\mathcal{P}$ uses the exterior-form indices of the curvature two-forms. The internal-dual contraction $\varepsilon^{abcd}R_{abmn}R_{cd}{}^{mn}$ is a distinct scalar. Equation (D-06) identifies the boundary currents explicitly and separates the local curvature contributions from their integrated characteristic densities.

3.4 Periodic stationary branches and their second variation

Proposition (positive coupling). On a connected circle of physical length L, all smooth real periodic stationary torsion profiles of the Ricci-tracefree action at fixed geometry and $\alpha > 0$ are the branches below. Here $z = Ly$, the two signs are independent, and $a^2 + b^2 > 0$ on the harmonic branch.

$$\begin{aligned} H &= \frac{1}{4\kappa\sqrt{\alpha}}, \qquad \alpha_n = \frac{L^2}{16\pi^2\kappa^2 n^2}, \quad n \geq 1, \\ (h, V) &= (0,0) \quad \text{or} \quad (\pm H, \pm 3H), \\ V &= 0, \quad h = a\cos(2\pi nz/L) + b\sin(2\pi nz/L), \quad \alpha = \alpha_n. \end{aligned} \tag{D-07}$$

Proof. On the open set where $V \neq 0$, the algebraic equation (D-03) fixes $h^2 = H^2$. Smoothness makes h constant on each connected component, and the other Euler equation then fixes $|V| = 3H$. By continuity this open set is also closed: a boundary point cannot have $V = 0$. Connectedness therefore makes it the whole circle or the empty set. In the latter case $V \equiv 0$ and $\ddot{h} + h/(4\alpha\kappa^2) = 0$ gives the resonance condition in (D-07). This argument excludes smooth splicing of the algebraic branches without dividing by the coefficient of V.

For perturbations (ϕ, v) about $(\bar{h}, \bar{V})$, the bulk second variation is

$$\begin{aligned} \delta^2\ell = &\left(\frac{6}{\kappa^2} - \frac{32\alpha}{3}\bar{V}^2\right)\phi^2 + \left(\frac{2}{3\kappa^2} - \frac{32\alpha}{3}\bar{h}^2\right)v^2 \\ &- \frac{128\alpha}{3}\bar{h}\bar{V}\phi v - 24\alpha\dot{\phi}^2. \end{aligned} \tag{D-08}$$

The zero and harmonic branches have a positive constant axial direction and negative sufficiently high axial Fourier directions, already at $v = 0$. At resonance the sine and cosine axial directions are also zero directions. For a mixed constant branch every Fourier block has

$$H_m = \begin{pmatrix} -24\alpha\xi_m^2 & \pm 4/\kappa^2 \\ \pm 4/\kappa^2 & 0 \end{pmatrix}, \qquad \xi_m = \frac{2\pi m}{L}, \qquad \det H_m = -\frac{16}{\kappa^4}. \tag{D-09}$$

Thus all positive-coupling stationary branches are saddles of this restricted Euclidean functional. The harmonic branch has zero action difference from zero torsion; the mixed branch has difference per four-volume $3/(16\alpha\kappa^4) > 0$. For $\alpha \leq 0$, (D-02) is a sum of nonnegative torsion terms and zero torsion is its unique minimum. On this fixed-metric family the LC-Weyl and irreducible EC-Weyl actions have that unique torsion minimum for every α. The mixed constant solutions are algebraic consequences of Paper II's static term; the periodic completeness and second-variation statements provide the extension here.

4 Static transverse deformations

4.1 Independent normalized shape directions

Use the Frobenius-orthonormal traceless symmetric basis $\mathrm{tr}(T_A T_B) = \delta_{AB}$ and the single exponential chart $M = \exp(\sum_A q_A T_A)$:

$$\begin{aligned} T_1 &= \frac{\mathrm{diag}(1,1,-2)}{\sqrt{6}}, \qquad T_2 = \frac{\mathrm{diag}(1,-1,0)}{\sqrt{2}}, \\ T_3 &= \frac{E_{01}+E_{10}}{\sqrt{2}}, \quad T_4 = \frac{E_{02}+E_{20}}{\sqrt{2}}, \quad T_5 = \frac{E_{12}+E_{21}}{\sqrt{2}}. \end{aligned} \tag{H-01}$$

Here E_{ij} is the matrix unit. At fixed r, L, η, V, the aligned static torsion contribution is independent of M in all three models. The shape derivatives therefore come from $-R_{\mathrm{LC}}/(2\kappa^2) - \alpha I_{\mathrm{LC}}$. For an arbitrary constant unimodular M, the spatial structure matrix is $N = r^{-1} M N_0 M^{\mathsf{T}}$, where N_0 is respectively $4I$, 0, or $-E_{22}$. This symmetric matrix can be diagonalized to the n_i of Section 3, extending those identities to all five shape directions.

For the round sphere, rotational invariance makes the Hessian a multiple of the identity in this basis. A one-direction calculation with $M = \mathrm{diag}(t, t, t^{-2})$, $t = \exp(q_1/\sqrt{6})$, gives $R''_{\mathrm{LC}}(0) = -64/r^2$ and $I''_{\mathrm{LC}}(0) = 4096/r^4$. Consequently,

$$\nabla_q U_X|_{S^3,0} = 0, \qquad H_X^{\mathrm{shape}}\Big|_{S^3,0} = \left(\frac{8\pi^2 L r}{\kappa^2} - \frac{1024\pi^2 L\alpha}{r}\right) I_5. \tag{H-02}$$

For $\alpha = 0$, $r = 3$, $L = \kappa = 1$, this is $24\pi^2 I_5$, as in Paper III. Flat T^3 has zero shape gradient and Hessian on the entire constant volume-preserving family, including the aligned MX torsion term, which depends only on the fixed amplitudes and volume. A nonzero torsionful EC curvature on T^3 is compatible with these zero shape derivatives.

4.2 Nil: a nonstationary shape and five path coefficients

For Nil, $N = -(Me_2)(Me_2)^{\mathsf{T}}/r$ has rank one. Define $z = (M^{\mathsf{T}}M)_{22} > 0$. Its only nonzero eigenvalue is $-z/r$, so $R_{\mathrm{LC}} = -z^2/(2r^2)$ and $I_{\mathrm{LC}} = 4z^4/(3r^4)$. The full aligned static potential is

$$\begin{aligned} U_X &= Az^2 + Bz^4 + U_{\mathrm{tor},X}, \\ A &= \frac{c_{\mathrm{Nil}^3} L r}{4\kappa^2}, \qquad B = -\frac{4\alpha c_{\mathrm{Nil}^3} L}{3r}, \\ U_{\mathrm{tor},X} &= c_{\mathrm{Nil}^3} L r^3 \left[\frac{3\eta^2}{\kappa^2 r^2} + \frac{V^2}{3\kappa^2} - \chi_X \frac{16\alpha\eta^2 V^2}{3r^2}\right]. \end{aligned} \tag{H-03}$$

For $f(z) = Az^2 + Bz^4$, let $f' = f'(1) = 2A + 4B$ and $f'' = f''(1) = 2A + 12B$. Differentiating $z = (\exp(2\sum q_A T_A))_{22}$ gives the full five-direction result

$$\begin{aligned} \nabla_q U_X|_0 &= \left(-\frac{4f'}{\sqrt{6}}, 0, 0, 0, 0\right), \\ H_X^{\mathrm{shape}}\Big|_0 &= \mathrm{diag}\left(\frac{32A+128B}{3}, 0, 0, 4A+8B, 4A+8B\right). \end{aligned} \tag{H-04}$$

The off-diagonal entries of this five-dimensional Hessian vanish at the chart origin. At $\alpha = -\beta < 0$ and $\eta = V = 0$, Paper II's radial slice critical point is $r_* = 4\kappa\sqrt{\beta/3}$, where $A = B > 0$. Evaluating the derivatives transverse to that radius slice gives

$$\begin{aligned} \partial_r U_X &= 0, \qquad \partial_L U_X = \frac{2A}{L}, \qquad \partial_\eta U_X = \partial_V U_X = 0, \\ \nabla_q U_X &= (-4\sqrt{6}A, 0, 0, 0, 0), \qquad \nabla_\omega U_X = \nabla_\delta U_X = 0, \\ H_X^{\text{shape}} &= A\,\mathrm{diag}\left(\frac{160}{3}, 0, 0, 12, 12\right). \end{aligned} \tag{H-05}$$

The nonzero shape and circumference derivatives make this point stationary only on the radius slice. Its shape second derivatives consist of two zero entries, a twofold equal pair, and one remaining entry in the specified exponential chart. The T_2 and T_3 exponentials stay entirely in the determinant-one 0–1 block and keep $z = 1$ exactly, giving two exactly flat paths. Appendix E uses the integral harmonic-form lattice to distinguish compact metrics on these paths and identifies discrete sign reversals.

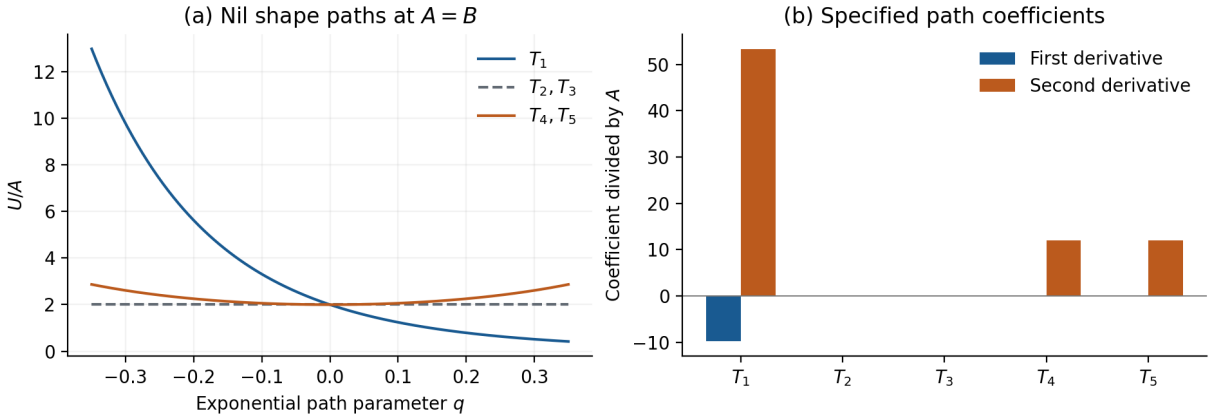


Figure 2: Nil shape paths and their derivatives, with $A = B = 1$ and zero torsion. The T_1 first derivative is nonzero at the reference point. The bars show first and second derivatives in the exponential chart (H-01). The two flat curves coincide exactly.

4.3 Static twist coefficients

In a diagonal spatial frame write $e^3 \mapsto e^3 + w_i e^i$. Then $C^3{}_{jk} = \sum_i w_i C^i{}_{jk}$, while $C^i{}_{jk}$ is unchanged. On this constant-twist family the tensor identity $W_{\text{EC}} = W_{\text{LC}}$ and $I_2 = I_3 = 8h^2V^2/3$ hold exactly. Hence the additional sectors have no twist derivatives in this family. For an undeformed spatial coframe, $w_i = L\omega_i/r$. At $\omega = 0$ the first derivative vanishes and all three actions have

$$\begin{aligned} H^\omega_{S^3} &= \frac{2\pi^2 L^3}{\kappa^2 r} I_3, \qquad H^\omega_{T^3} = 0, \\ H^\omega_{\text{Nil}^3} &= \mathrm{diag}\left(0, 0, \frac{c_{\text{Nil}^3} L^3}{2\kappa^2 r} - \frac{16\alpha c_{\text{Nil}^3} L^3}{3r^3}\right). \end{aligned} \tag{H-06}$$

The LC-Weyl twist Hessian is zero at round S^3; its invariant can start at higher order. Nil selects the noncommutative direction, whereas constant torus twists leave the local curvature invariants unchanged. These coefficients measure static directional costs. For the derivative

terms, Paper III finds a vanishing quadratic twist velocity term at zero twist when $\alpha = 0$, and Section 6 evaluates a coupled shape–twist family at nonzero coupling.

For completeness, at $q = \omega = \delta = \eta = V = 0$ the round-sphere background has $\partial_r U = -12 c_{S^3} L/\kappa^2$ and $\partial_L U = -12 c_{S^3} r/\kappa^2$. The round-sphere coefficients above are therefore coefficients along the displayed directions at a point with nonzero radius and circumference derivatives. At the torsion-free flat-torus point the displayed restricted potential and its first derivatives vanish.

5 Torsion-axis orientation and interaction order

5.1 The extra nine-component sector sees anisotropy

Keep the product metric and the n_i constant, set twist to zero, and let u be a constant unit vector in the unrotated four-frame. Respecifying aligned torsion after a constant coframe rotation is equivalent to

$$T^{\mathrm{AX}}_{abc} = 2h\varepsilon_{abcd}u_d, \qquad T^{\mathrm{VT}}_{abc} = \frac{V}{3}(\delta_{ac}u_b - \delta_{ab}u_c), \qquad u_a u_a = 1. \tag{O-01}$$

The exact static result is

$$\begin{aligned} W_{\mathrm{EC}} &= W_{\mathrm{LC}}, \qquad R_{\mathrm{EC}} = R_{\mathrm{LC}} - 6h^2 - \frac{2}{3}V^2, \\ I_2 &= \frac{8}{3}h^2V^2 + 4h^2\sum_{i=0}^{2}(n_j - n_k)^2 u_i^2, \\ I_3 &= \frac{8}{3}h^2V^2, \qquad \mathcal{N} = \mathcal{P} = 0. \end{aligned} \tag{O-02}$$

In the sum, (i, j, k) is cyclic. Only the extra nine-component sector produces this orientation dependence. It vanishes at round S^3 and flat T^3, but need not vanish on an anisotropic shape or on Nil. The scalar, LC-Weyl, and irreducible EC-Weyl actions are independent of this constant orientation. Equation (O-02) is an identity for constant orientations u.

For the ordered mixing rotation $O = R_{03}(\delta_0)R_{13}(\delta_1)R_{23}(\delta_2)$, with $\tan\vartheta_i = \delta_i$, put $c_i = (1+\delta_i^2)^{-1/2}$ and $s_i = \delta_i c_i$. The torsion axis is $u = O^{\mathsf{T}} e_3$, with

$$u = (-s_0, -c_0 s_1, -c_0 c_1 s_2, c_0 c_1 c_2). \tag{O-03}$$

Each u_i^2 is even in every δ_j. Thus all pure cubic mixing monomials vanish at $\delta = 0$ on this family. The quadratic orientation coefficients in the trace-free model are

$$H^{\delta}_{\mathrm{tf}} = -8\alpha c_M L r^3 h^2 \,\mathrm{diag}((n_1 - n_2)^2, (n_0 - n_2)^2, (n_0 - n_1)^2). \tag{O-04}$$

At the undeformed Nil shape this gives equal δ_0, δ_1 coefficients and a zero δ_2 coefficient. Expanding around a nonzero $\bar{h}$ with $h = \bar{h} + \varphi$ produces the mixed cubic term

$$\Delta U_{\mathrm{tf}}|_{\varphi\delta^2, \mathrm{Nil}^3} = -\frac{8\alpha c_{\mathrm{Nil}^3} L r^3 \bar{h}}{r^2}\,\varphi(\delta_0^2 + \delta_1^2). \tag{O-05}$$

This interaction couples an axial-amplitude fluctuation to two orientation fluctuations. It belongs entirely to the extra pair-antisymmetric curvature sector. Pure orientation cubics vanish by parity, while the background axial amplitude permits this mixed cubic.

5.2 Field order of the aligned mixed term

The aligned static term $-16\alpha h^2V^2/3$ from Paper II is fourth order in the field amplitudes at zero torsion. Its expansion provides a comparison with the orientation interaction above. Let $h = \bar{h} + \varphi$ and $V = \bar{V} + v$. The degree-two, degree-three, and degree-four pieces of h^2V^2 are, respectively,

$$\begin{aligned} [h^2V^2]_2 &= \bar{V}^2\varphi^2 + 4\bar{h}\bar{V}\varphi v + \bar{h}^2v^2, \\ [h^2V^2]_3 &= 2\bar{V}\varphi^2 v + 2\bar{h}\varphi v^2, \\ [h^2V^2]_4 &= \varphi^2 v^2. \end{aligned} \tag{O-06}$$

At a nonstationary background the constant and linear pieces also enter the expansion. Equation (O-06) expresses the field order of Paper II's static potential in terms of the fluctuations φ, v, with α held fixed. The cubic terms therefore depend on the chosen background; field order and derivative order label different features of the interaction.

5.3 Finite-angle extrema and degeneracies

Equation (O-02) also determines the global orientation extrema at fixed geometry and amplitudes. On the unit three-sphere,

$$\begin{aligned} \Delta U_{\mathrm{tf}} &= K_u u^{\mathsf{T}} D u, \qquad K_u = -4\alpha\mathcal{V}h^2, \\ D &= \mathrm{diag}((n_1-n_2)^2, (n_0-n_2)^2, (n_0-n_1)^2, 0), \qquad \mathcal{V} = c_M L r^3. \end{aligned} \tag{O-07}$$

For $K_u \neq 0$, Lagrange multipliers give $Du = \lambda u$. Hence every critical set is the unit sphere in an eigenspace E_λ of D, with tangent Hessian

$$u \in S(E_\lambda), \qquad H_{\tan} = 2K_u(D - \lambda I)|_{u^\perp}\,. \tag{O-08}$$

For $\alpha < 0$ the minimum is $S(\ker D)$ and the maximum is the sphere of the largest eigenspace; for $\alpha > 0$ they exchange. Intermediate eigenspheres are saddles. The degeneracies are exhaustive: if the three distinct n_i are ordered with consecutive gaps $a, b > 0$, the eigenvalues are $0, a^2, b^2, (a+b)^2$. Unequal gaps give four antipodal pairs; equal gaps give an intermediate critical circle of saddles. Exactly two equal n_i give two critical circles, with eigenvalues $0, 0, d^2, d^2$. All equal n_i, $\alpha = 0$, or $h = 0$ make the whole orientation sphere flat.

On Nil, $D = r^{-2}\mathrm{diag}(1, 1, 0, 0)$. For $\alpha < 0$ its minimum circle lies in $\mathrm{span}(e_2, e_3)$ and its maximum circle in $\mathrm{span}(e_0, e_1)$, each with one tangent zero direction and two equal transverse eigenvalues. For $\alpha \leq 0$ the orientation contribution is nonnegative at every shape and vanishes at the aligned axis $u = e_3$. Allowing this constant-axis freedom therefore does not lower Paper II's aligned static infimum over the same shape domain and fixed controls. This is a corollary of (O-02).

6 Coupled periodic shape and twist

6.1 Curvature components with metric derivatives

Keep r, L, η, V fixed on $S^3 \times S^1$, with $h = \eta/r$ and torsion aligned with the third frame direction. Consider the coframe below and, for the frequency sequence, Paper III's nondegenerate family (X4):

$$\begin{gathered} e^0 = rf\sigma^0, \quad e^1 = rf^{-1}\sigma^1, \quad e^2 = r\sigma^2, \quad e^3 = L(\mathrm{d}y + w\sigma^0), \quad f = e^s, \\ w_n(y) = A\cos(2\pi n y), \qquad f_n^2 = 1 - \frac{L^2 w_n^2}{2r^2}, \qquad 0 < A < \frac{\sqrt{2}r}{L}, \quad n \geq 1. \end{gathered} \tag{J-01}$$

First allow arbitrary smooth $f(y) > 0, w(y)$ in this coframe. Primes denote y derivatives, not the physical-circle derivative used in Section 3. Keeping the first and second coframe derivatives in the Cartan curvature gives

$$\begin{gathered} W_{\mathrm{EC}} = W_{\mathrm{LC}}, \qquad R_{\mathrm{EC}} - R_{\mathrm{LC}} = -6h^2 - \frac{2}{3}V^2, \qquad \mathcal{N} = 0, \\ I_2 = \frac{8}{3}h^2V^2 + 4h^2\left(\frac{(w')^2}{f^2r^2} + \frac{4(s')^2}{L^2}\right), \qquad I_3 = \frac{8}{3}h^2V^2. \end{gathered} \tag{J-02}$$

One can obtain the torsion-dependent part without expanding the entire metric curvature. With $A^a{}_\mu$ the coframe matrix relative to $(\sigma^0, \sigma^1, \sigma^2, \mathrm{d}y)$, let $B = A^{-1}$, $b_a = B^y{}_a$, and $D = A'B$. The derivative contribution to the structure tensor is $(C_\partial)^a{}_{bc} = D^a{}_c b_b - D^a{}_b b_c$, in addition to its static part. Here $b = (-w/(rf), 0, 0, 1/L)$, $D_{00} = s'$, $D_{11} = -s'$, and $D_{30} = Lw'/(rf)$ are the nonzero entries. The frame components of contortion are constant. Substituting these quantities into (G-04) and the curvature formula (B-02), the torsion difference has zero Weyl projection. Its derivative part lies in the extra nine-component sector, is orthogonal to the static mixed part, and has norm $4h^2[(w')^2/(f^2r^2) + 4(s')^2/L^2]$. The four-form part remains constant. Both contractions entering complete NY equal $-4hV$, giving its pointwise cancellation in (J-02).

The constant $16h^2V^2/3$ in $I_{\mathrm{tf}} - I_{\mathrm{LC}}$ is Paper II's static result. The derivative term in (J-02) is the additional contribution from the varying geometry; it is carried entirely by the nine-component sector.

6.2 Frequency polynomial and strict quartic positivity

For (J-01), write coefficient functions at the base phase ny modulo one. Structure and connection have degree one in n, so curvature has degree two. The static coefficients are even under phase reflection, first-derivative coefficients are odd, and second-derivative coefficients are even. Odd powers therefore vanish upon integration over a period. Define

$$\begin{gathered} \mathcal{R}(n) = \mathcal{R}_0 + n\mathcal{R}_1 + n^2\mathcal{R}_2, \\ \int_0^1 I_{\mathrm{LC}}(n)\,\mathrm{d}y = Q_0 + Q_2 n^2 + Q_4 n^4, \qquad Q_4 = \int_0^1 \|\Pi_W \mathcal{R}_2\|^2\,\mathrm{d}y, \\ D_A = 4h^2\int_0^1 \left(\frac{(w_1')^2}{f_1^2 r^2} + \frac{4(s_1')^2}{L^2}\right)\mathrm{d}y. \end{gathered} \tag{J-03}$$

Here Π_W denotes the Weyl projection of Appendix A. The frequency coefficient $\mathcal{R}_2$ in (J-03) is distinct from the irreducible sector $R^{(2)}$. It is independent of the constant contortion and has LC

Riemann symmetries. Consequently all three quadratic invariants share the same Q_4. Moreover, it is strictly positive for every fixed admissible nonzero A. At $y = 1/4$, one has $w_1 = 0, w_1'' = 0$, $s_1' = 0$, and $s_1'' = -L^2v^2/(2r^2)$. The only independent nonzero components of $\mathcal{R}_2$ there are $(\mathcal{R}_2)_{0303} = (\mathcal{R}_2)_{1313} = -v^2/(2r^2)$, which give

$$\|\Pi_W \mathcal{R}_2\|^2\Big|_{y=1/4} = \frac{v^4}{3r^4} = \frac{(2\pi A)^4}{3r^4} > 0, \qquad v = w_1'(1/4) = -2\pi A. \tag{J-04}$$

Since the integrand defining Q_4 is nonnegative and smooth, (J-04) proves $Q_4 > 0$, rather than merely checking selected amplitudes. Also $D_A \geq 0$, with strict positivity for $h \neq 0$.

Let $\overline{U}_A$ be the mean static EC action along this path, without the curvature-squared term. Paper III supplies its scalar derivative coefficient C_A. The exact integrated actions for integer $n \geq 1$ are

$$\begin{aligned} S_{\mathrm{LC}}(n) = S_{\mathrm{W}}(n) &= \overline{U}_A - C_A n^2 - \alpha\mathcal{V}(Q_0 + Q_2 n^2 + Q_4 n^4), \\ S_{\mathrm{tf}}(n) - S_{\mathrm{W}}(n) &= -\alpha\mathcal{V}\left(\frac{16}{3}h^2V^2 + D_A n^2\right), \\ C_A &= \frac{\pi^2 L^3}{16\kappa^2 r}\int_0^1 \frac{w_1^2 (w_1')^2}{f_1^4}\,\mathrm{d}y > 0, \qquad \mathcal{V} = \frac{\pi^2 L r^3}{4}. \end{aligned} \tag{J-05}$$

At fixed nonzero amplitude, all three actions have leading term $-\alpha\mathcal{V}Q_4 n^4$. They tend to $-\infty$ for $\alpha > 0$ and to $+\infty$ for $\alpha < 0$. At $\alpha = 0$ they return to Paper III's $\overline{U}_A - C_A n^2 \to -\infty$. Thus negative coupling removes the high-frequency descent on this particular fixed-amplitude sequence. The statement concerns integer frequencies at fixed amplitude A.

6.3 A finite-frequency comparison

For $r = 3$, $L = \kappa = \eta = V = 1$, and $A = 0.5$, the coefficients are

$$Q_0 = 0.0003213884274, \quad Q_2 = 0.01228429824, \quad Q_4 = 0.2062093292, \quad D_A = 0.2462621564.$$

The scalar part has $\overline{U}_A = -44.0019357894$ and $C_A = 0.06430979322$, reproducing Paper III at zero coupling. At $\alpha = -10^{-4}$ the minimum over positive integer frequencies of this fixed-amplitude sequence is at $n = 5$ in all three models. The LC-Weyl and irreducible EC-Weyl actions are -44.7490306345, compared with the round reference -44.4132198049. The Ricci-tracefree action is -44.7040679354, compared with its own round reference -44.4092719631. Positive quartic growth thus coexists with lower-action configurations at finite frequency. Both are comparisons of the action along the prescribed family (J-01).

7 Conclusions and scope of the results

Paper II's curvature decomposition distinguishes the restricted torsion variations of the three quadratic-curvature actions. At fixed product geometry, the nine-component and four-form sectors carry the axial derivative term, while irreducible EC-Weyl remains equal to LC-Weyl. For the Ricci-tracefree action, nonpositive coupling admits only zero smooth periodic stationary torsion. At positive coupling the complete classification consists of zero, resonant harmonic and mixed constant branches, all saddles of the restricted functional. Local Nieh–Yan and Pontryagin densities are boundary currents with vanishing periodic integrals.

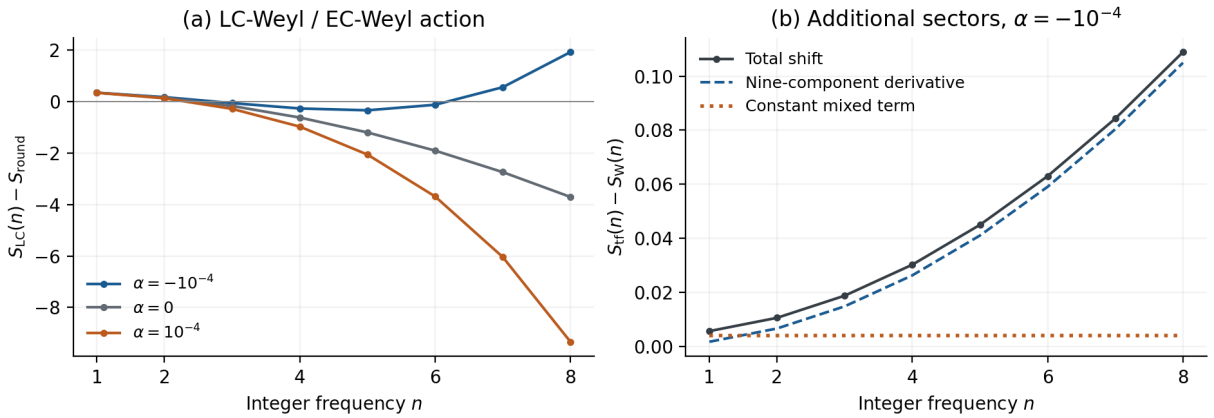


Figure 3: Coupled profiles (J-01) at $r = 3$, $L = \kappa = \eta = V = 1$, $A = 0.5$. Left: LC-Weyl and irreducible EC-Weyl action relative to the round reference, for $\alpha = -10^{-4}, 0, 10^{-4}$. Right: the Ricci-tracefree action shift at $\alpha = -10^{-4}$, separated into the inherited constant mixed term and the nine-component derivative contribution. Frequencies are positive integers.

Constant torsion-axis orientations reduce to eigenspheres of an anisotropy matrix, giving their finite-angle extrema and every degeneracy. For nonpositive coupling, releasing this constant-axis freedom does not lower the aligned static infimum of Paper II. Five normalized shape directions give the Nil gradient and flat paths in a common chart. The nonzero shape and circumference derivatives remain, while an integral harmonic-form invariant distinguishes continuously many compact metrics along those flat paths. Appendix C explains the gradient term in coordinate transformations of nonstationary Hessians.

On the specified coupled shape–twist family, the extra nine-component sector alone supplies the additional metric-derivative term. All three actions share a strictly positive quartic curvature coefficient. Consequently the fixed-amplitude high-frequency actions grow for negative coupling, descend without bound for positive coupling, and recover Paper III's quadratic descent at zero coupling. Finite-frequency configurations can still have lower action than the round reference when the quartic growth is positive.

These results hold for the specified Euclidean variation families. Within those domains, the component formulas separate the inherited static contributions from the torsion-gradient, orientation, and coupled metric-derivative effects. Extending the analysis to general joint profiles and to the field and constraint equations of the full theory is left for future work.

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Availability and contact

The manuscript source and supplement are associated with https://github.com/Muacca/DPPUv2-paper03ec and the archival record 10.5281/zenodo.19425147. Use the version information in the manuscript and dataset manifest when selecting files; earlier releases use different formulas and data. Contact: muacca@dmwp.jp; issues may be reported through the repository.

A Curvature projections and LC normalization

For a metric-compatible curvature array, $R_{abcd} = -R_{bacd} = -R_{abdc}$. Define $Q_{bd} = \sum_a R_{abad}$ and $R = \sum_b Q_{bb}$. Do not impose pair exchange or symmetric Ricci. The raw Ricci subtraction is the linear map

$$\begin{aligned}(\Pi R)_{abcd} = R_{abcd} &- \frac{1}{2}(\delta_{ac}Q_{bd} - \delta_{ad}Q_{bc} - \delta_{bc}Q_{ad} + \delta_{bd}Q_{ac}) \\ &+ \frac{R}{6}(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}).\end{aligned} \tag{A-01}$$

Let S denote pair symmetrization and F its four-form part. The three required projections are

$$\begin{aligned}S_{abcd} = \tfrac{1}{2}(R_{abcd} + R_{cdab}), \qquad F_{abcd} = \tfrac{1}{3}(S_{abcd} + S_{acdb} + S_{adbc}), \\ W = \Pi(S - F), \qquad R^{(3)} = F, \qquad R^{(2)} = \Pi R - W - F.\end{aligned} \tag{A-02}$$

The trace of F is zero. $S - F$ has the pair symmetry and cyclic identity of an algebraic Riemann tensor, so removing its Ricci part leaves the ten-dimensional Weyl component. In the 6×6 matrix of antisymmetric index pairs, the pair-antisymmetric subspace has dimension fifteen; its six-dimensional antisymmetric Ricci part is removed by Π, leaving nine components. The four-form subspace has dimension one. These mutually orthogonal subspaces account for the twenty-dimensional Ricci-free image. The projectors are self-adjoint and idempotent in the Euclidean sum-over-all-indices inner product; hence their squared norms add as in (C-01). The supplementary calculation also constructs the linear maps on all thirty-six pair-matrix basis elements.

For a product of a homogeneous three-geometry and a flat circle, diagonal structure constants n_i give spatial Ricci eigenvalues

$$\begin{aligned}\lambda_i = \frac{1}{2}[n_i^2 - (n_j - n_k)^2], \qquad R_{\mathrm{LC}} = \sum_i \lambda_i, \\ I_{\mathrm{LC}} = 2\sum_i \lambda_i^2 - \frac{2}{3}R_{\mathrm{LC}}^2.\end{aligned} \tag{A-03}$$

The last equality follows by inserting the three-dimensional identity $\|\mathrm{Riem}_3\|^2 = 4\|\mathrm{Ric}_3\|^2 - R_3^2$ into the four-dimensional Weyl norm. It gives $R_{\mathrm{LC}} = 24/r^2$, $I_{\mathrm{LC}} = 0$ for the round sphere, both zero for the torus, and $R_{\mathrm{LC}} = -1/(2r^2)$, $I_{\mathrm{LC}} = 4/(3r^4)$ for undeformed Nil. These values set the normalization of all tables.

B Derivative calculation and component bookkeeping

For cyclic spatial structure constants and aligned torsion, the nonzero connection coefficients, together with antisymmetry in the first two indices, are

$$\Gamma_{ijk} = \left(\frac{n_i + n_j - n_k}{2} - h\right)\varepsilon_{ijk}, \qquad \Gamma_{i3j} = -\frac{V}{3}\delta_{ij}, \qquad \Gamma_{3ij} = \frac{V}{3}\delta_{ij}. \tag{B-01}$$

The full noncoordinate-frame curvature is

$$R_{abcd} = D_c\Gamma_{abd} - D_d\Gamma_{abc} + \Gamma_{aec}\Gamma_{ebd} - \Gamma_{aed}\Gamma_{ebc} + \Gamma_{abe}C^{e}{}_{cd}. \tag{B-02}$$

The first two terms retain the derivatives of the connection, including contortion. For circle profiles, projection of (B-02) gives the LC Weyl tensor and the following additional entries:

$$\begin{aligned} a_+ &= \tfrac{1}{2}\left(\dot{h} + \tfrac{2}{3}hV\right), \qquad a_- = \tfrac{1}{2}\left(\dot{h} - \tfrac{2}{3}hV\right), \\ R^{(2)}_{ijk3} &= a_+\varepsilon_{ijk}, \qquad R^{(2)}_{k3ij} = -a_+\varepsilon_{ijk}, \\ R^{(3)}_{abcd} &= a_-\varepsilon_{abcd}. \end{aligned} \tag{B-03}$$

All other entries follow from antisymmetry within pairs, or vanish. Summing the twenty-four nonzero signed entries in each component gives $24a_+^2$ and $24a_-^2$, proving (D-01). Contracting the unprojected curvature gives the scalar term in (D-01); contracting its four-form part gives $12a_- = 6\dot{h} - 4hV$ in (D-06). A separate exterior-algebra construction verifies the first Cartan equation, both differential Bianchi identities, and the Nieh–Yan identity.

The Pontryagin density in (D-06) follows either from the exterior product of the curvature two-forms or from the connection Chern–Simons form. With the matrix-trace convention, $\mathrm{Tr}(R \wedge R) = -\sum_{a,b} R_{ab} \wedge R_{ab}$, so $\mathrm{dCS}(\omega) = -\mathcal{P}\,\mathrm{vol}$. This also fixes the sign of the displayed primitive.

To expose the cancellation between the two sectors, introduce separate symbols b_2, b_3 only as bookkeeping weights. Their bulk contribution would read

$$-b_2 I_2 - b_3 I_3 = -6(b_2 + b_3)\dot{h}^2 - 8(b_2 - b_3)hV\dot{h} - \frac{8}{3}(b_2 + b_3)h^2V^2. \tag{B-04}$$

The action compared in this paper has $b_2 = b_3 = \alpha$, not independent physical couplings. Moreover, $I_2 - I_3 = 16hV\dot{h}$ need not integrate to zero when V varies. In physical circle coordinate z,

$$\int_0^L (I_2 - I_3)\,\mathrm{d}z = -8\int_0^L \dot{V}h^2\,\mathrm{d}z. \tag{B-05}$$

The cancellation in (D-02) is therefore a consequence of equal weights, not a general claim that the two component functionals are identical. The reduction to the scalar action at $\alpha = 0$ and the static Paper II limit at $\dot{h} = \dot{V} = 0$ are separate checks.

C Coordinate maps, gradients, and variation paths

At a nonstationary point the second derivative changes with the second derivative of the coordinate map. If $q = q(x)$, then

$$\frac{\partial^2 U}{\partial x^a \partial x^b} = \frac{\partial q^i}{\partial x^a} H_{ij} \frac{\partial q^j}{\partial x^b} + \frac{\partial U}{\partial q^i} \frac{\partial^2 q^i}{\partial x^a \partial x^b}. \tag{P-01}$$

C.1 Nil coordinate redundancy

Consider the Nil diagonal parametrization $M = \mathrm{diag}(1, f, f^{-1})$, $f = (1+\varepsilon)^{2/3}(1+s)$. Both parameters change the same diagonal shape. In the normalized basis (H-01),

$$q_1 = \frac{\sqrt{6}}{2} \log f, \qquad q_2 = -\frac{1}{\sqrt{2}} \log f, \qquad q_3 = q_4 = q_5 = 0, \qquad \mathrm{rank} \frac{\partial(q_1, q_2)}{\partial(\varepsilon, s)} = 1. \tag{P-02}$$

The corresponding invariant is $z = f^{-2}$. Direct differentiation of $Az^2 + Bz^4$, or transformation by (P-01) including the gradient term, gives

$$H_{\varepsilon,s} = \begin{pmatrix} \frac{88A+304B}{9} & \frac{32A+128B}{3} \\ \frac{32A+128B}{3} & 20A + 72B \end{pmatrix}. \tag{P-03}$$

The gradient term allows this matrix to have rank two even though the tangent map has rank one. Its eigenvalues consequently describe this parametrized family; the five independent shape directions are those of (H-01). In particular, the cross coefficient is

$$\partial_\varepsilon \partial_s U_X|_0 = \frac{128\pi^4 L r}{3\kappa^2} - \frac{8192\pi^4 L\alpha}{9r}. \tag{P-04}$$

The first term comes from EC scalar curvature and the second from the LC-Weyl invariant, equal here to irreducible EC-Weyl. The additional two Ricci-free sectors are shape independent on this aligned family. Equation (P-04) thus separates the two geometric contributions to the cross coefficient on this path.

C.2 Sphere and torus path dictionary

The sphere diagonal chart is $M = \mathrm{diag}(a(1+s), a/(1+s), a^{-2})$, $a = (1+\varepsilon)^{1/3}$. It has

$$q_1 = \frac{\sqrt{6}}{3} \log(1+\varepsilon), \qquad q_2 = \sqrt{2} \log(1+s). \tag{P-05}$$

At the round point the shape gradient vanishes; if H_0 is the coefficient in (H-02), the diagonal-chart entries are $H_{\varepsilon\varepsilon} = 2H_0/3$, $H_{ss} = 2H_0$, and $H_{\varepsilon s} = 0$. Together with the global volume factor, this map gives the normalization in each set of coordinates.

For the torus, compare a volume-changing path with volume factor $(1+\varepsilon)(1+s)$, keeping r, L, η, V fixed. With the period-one circle normalization, this path gives

$$\begin{aligned} \partial_\varepsilon \partial_s U_X^{\text{volume path}}\Big|_0 &= c_{T^3} L r^3 \left[\frac{3\eta^2}{\kappa^2 r^2} + \frac{V^2}{3\kappa^2} - \chi_X \frac{16\alpha\eta^2 V^2}{3r^2} \right], \\ \partial_\varepsilon \partial_s U_X^{\text{volume path}}\Big|_{V=0,0} &= \frac{24\pi^3 L r \eta^2}{\kappa^2}. \end{aligned} \tag{P-06}$$

The corresponding cross term is zero on the volume-preserving exponential shape family. This difference is due to the variation of volume. Although LC curvature vanishes on the torus, the

EC scalar $R_{\mathrm{EC}} = -6h^2 - 2V^2/3$ and static trace-free correction $16h^2V^2/3$ can be nonzero and contribute along the volume-changing path.

D Reproduction and revision note

D.1 Code and data

The supplement contains the current Cartan and projection formulas, an exterior-algebra calculation, exact symbolic results, and the data for the three figures. It requires Python 3.11 or later and the dependencies in `script/requirements.txt`; the notebook additionally requires a Jupyter environment. Recomputing the results and rendering the saved results are separate operations. From the `script` directory:

```
python -m pip install -r requirements.txt
python -B paper03ec.py verify
python -B paper03ec.py recompute --output ../recomputed
python -B paper03ec.py render --data ../data/paper03ec_v4 --output ../rendered
```

Output directories must be new. The notebook `DPPUv2_Paper_Figures_EC.ipynb` loads saved data by default through its adjacent viewer; recalculation is an explicit option that writes to a new time-stamped folder. The manifest records the prescribed profiles, coordinate chart, conventions, and file hashes. The symbolic calculation reproduces the tensor identities, scalar reductions, static transverse formulas, and Euler equations, the stationary-branch coefficients, the coupled-profile frequency terms, and the Nil harmonic-lattice formulas. Figure rendering uses the saved numerical data independently of that calculation.

D.2 Revision note

Version 4 adopts the complete-NY curvature coefficient $1/2$ and the global normalization of Papers I–II. Earlier computational inputs used $T_{\mathrm{geom}} = -\tau$; the present supplement uses geometric torsion directly. The curvature comparison distinguishes irreducible EC-Weyl from Ricci-tracefree EC curvature and extends Paper II's static identities to the variations specified here.

The preceding universal Palatini-protection and unchanged spin-2 kinetic claims, the full homogeneous Nil minimum and its $|\kappa^2\theta| = 1$ threshold, and the particle-mass and spin-1 tachyon interpretations are withdrawn. Sections 3–5 give the restricted equations and deformation coefficients in their place. The internal-dual Pontryagin interpretation and the associated topological and chiral conclusions are also withdrawn. Appendix C gives the coordinate and path comparisons; the repository's `CHANGELOG.md` records the detailed changes to the preceding version, including interaction order and normalization.

E Flat Nil paths on the fixed compact quotient

The paths T_2, T_3 lie in the positive determinant-one horizontal metrics below, with the fixed coordinate periods and gluing of Section 2. Their shape variable $(M^{\mathsf{T}}M)_{22}$ remains one, so (H-03) is constant along them.

$$
\begin{aligned}
g_Q &= r^2\left[(\mathrm{d}x,\mathrm{d}w)Q(\mathrm{d}x,\mathrm{d}w)^{\mathsf{T}} + (\mathrm{d}z - x\mathrm{d}w)^2\right] + L^2\mathrm{d}y^2,\\
Q &> 0, \qquad \det Q = 1, \qquad G_{H^1} = \frac{\mathcal{V}}{4\pi^2 r^2}Q^{-1}.
\end{aligned}
\tag{N-01}
$$

The last expression is the L^2 Gram matrix of the integral one-forms $\mathrm{d}x/(2\pi), \mathrm{d}w/(2\pi)$, with $\mathcal{V} = (2\pi)^4 L r^3$. These forms are closed and coclosed: the horizontal block of the inverse coordinate metric is $r^{-2}Q^{-1}$ and the volume density is constant, so their divergences vanish. The abelianization of the Nil lattice has rank two; these forms, together with $\mathrm{d}y$ on the product, span harmonic degree one. The LC Ricci eigenvalues are $-1/(2r^2)$ horizontally, $+1/(2r^2)$ centrally, and zero on the circle. Any product-metric isometry therefore preserves the horizontal harmonic subspace and its integral lattice. At fixed r, L, its normalized shortest squared norm is an isometry invariant:

$$
\begin{aligned}
\sigma(Q) &= \min_{m\in\mathbb{Z}^2\setminus\{0\}} m^{\mathsf{T}}Q^{-1}m, \qquad a = \sqrt{2}q,\\
Q_{T_2} &= \mathrm{diag}(e^a, e^{-a}), && \sigma(Q_{T_2}) = e^{-|a|},\\
Q_{T_3} &= \begin{pmatrix} \cosh a & \sinh a \\ \sinh a & \cosh a \end{pmatrix}, && \sigma(Q_{T_3}) = \min(\cosh a, 2e^{-|a|}).
\end{aligned}
\tag{N-02}
$$

For T_2, the smaller diagonal coefficient bounds every nonzero integer vector and is attained by a coordinate vector. For T_3, integer vectors of Euclidean squared length one give $\cosh a$; all others have squared length at least two and hence norm at least $2e^{-|a|}$, attained by $(1, \pm 1)$ with the appropriate sign. This proves (N-02) on the infinite lattice.

Distinct $|q|$ are distinguished on the entire T_2 path and on the interval $0 \le \sqrt{2}|q| < \log(3)/2$ of T_3. In that interval, nonzero T_2 points have $\sigma < 1$ and nonzero T_3 points have $\sigma > 1$, so the paths are also distinguished from each other there. Opposite signs are nevertheless identified by explicit lattice-preserving maps. In the normalized coordinates $(X, Y, Z) = (x/(2\pi), w/(2\pi), z/(4\pi^2))$, these are

$$
(X, Y, Z) \mapsto (Y, -X, Z - XY) \quad (T_2), \qquad (X, Y, Z) \mapsto (-X, Y, -Z) \quad (T_3). \tag{N-03}
$$

They preserve the integer Heisenberg group law and the square of $\mathrm{d}Z - X\mathrm{d}Y$, while taking $Q(q)$ to $Q(-q)$. Horizontal determinant-one transformations give local Heisenberg automorphisms, but need not preserve the lattice. This local/global distinction is consistent with the geometry of fixed Heisenberg quotients[4]. Here the integral harmonic norm serves as an obstruction to isometry. The flat paths contain continuously distinct compact metrics, along which the transverse derivatives in (H-05) remain nonzero.

References

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